MediCheck

API Contract — Service 1 to Service 2

Sprint 2 | Inter-Service Alignment Document | For team review before coding begins

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Purpose

This document defines the request and response contract for every endpoint that Service 1 (React frontend) calls on Service 2 (Bill Analysis API). Both team members must agree on all decisions in this document before writing any code that crosses the service boundary.

Once agreed, this document becomes the source of truth for the inter-service schema. Any changes must be proposed via pull request and approved by both members.

Endpoint 1 — POST /upload

The frontend sends one or two PDF files. Service 2 runs OCR extraction and returns a session identifier and the extracted fields for display on the field confirmation page.

Request

Content-Type: multipart/form-data

Field	Type	Required	Description
bill	PDF file	Yes	Provider bill PDF
eob	PDF file	No	Explanation of Benefits PDF

Response — HTTP 200 (success)

```
{ "session_id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "extracted",
  "extracted_fields": {
    "patient_name": "James Whitfield",
    "provider_name": "Atrium Health Carolinas Medical Center",
    "date_of_service": "2025-09-17",
    "total_billed": 12085.00,
    "line_items": [
      { "line_number": 1, "cpt_code": "29881", "description": "",
        "quantity": 1, "amount": 3200.00, "confidence": 0.97,
        "source": "bill" }
    ]
  },
```

```
  "rag_available": true }
```

Response — HTTP 400 (validation error)

```
{ "error_code": "INVALID_FILE_TYPE",
  "message": "Uploaded file is not a PDF.",
  "session_id": null }
```

Decisions to confirm

- **CPT descriptions:** Does description stay empty to avoid AMA copyright issues? Recommendation: Yes — never populated
- **Confidence threshold:** What confidence threshold highlights a field in yellow in the UI? Recommendation: 0.80 (80%)
- **Field separation:** Are bill and EOB fields returned separately with a source field? Recommendation: Yes — source: 'bill' or 'eob' on each line item. Separately so the UI can label which document each field came from

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Endpoint 2 — POST /confirm

The frontend sends the session identifier and all user-corrected field values after the field confirmation step. Service 2 persists the confirmed data and updates the session status.

Request

```
{ "session_id": "550e8400-e29b-41d4-a716-446655440000",
  "confirmed_fields": {
    "patient_name": "James Whitfield",
    "provider_name": "Atrium Health Carolinas Medical Center",
    "date_of_service": "2025-09-17",
    "total_billed": 12085.00,
    "line_items": [
      { "line_number": 1, "cpt_code": "29881",
        "quantity": 1, "amount": 3200.00, "source": "bill" }
    ]
  } }
```

Response — HTTP 200 (success)

```
{ "session_id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "confirmed" }
```

Response — HTTP 404 (session not found)

```
{ "error_code": "SESSION_NOT_FOUND",
  "message": "No session found for the provided session_id.",
```

```
  "session_id": "550e8400-e29b-41d4-a716-446655440000" }
```

Decisions to confirm

- **Should confidence scores be stripped from confirmed fields before sending, or passed through?**
  Recommendation: strip them — confidence is an extraction quality signal, not a confirmed data field

- **Line item structure:** Are line items sent as a flat array or grouped by document (bill vs EOB)? Recommendation: flat array with a source field on each item ("source": "bill" or "source": "eob")

Endpoint 3 — POST /analyse

The frontend triggers error detection after field confirmation. Service 2 runs all four detection checks, calls Service 3 for RAG explanations, and returns the complete results payload.

Request

```
{ "session_id": "550e8400-e29b-41d4-a716-446655440000" }
```

Response — HTTP 200 (success, RAG available)

```
{ "session_id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "analysed",
  "total_errors": 2,
  "total_estimated_savings": 747.00,
  "all_clear": false,
  "rag_available": true,
  "errors": [
    { "error_id": "err_001",
      "module": "duplicate_charge",
      "error_type": "Duplicate Charge",
      "description": "CPT 29881 appears twice on 2025-09-17.",
      "line_items_affected": [1, 7],
      "estimated_dollar_impact": 480.00,
      "confidence": "high",
      "explanation": "Duplicate billing for the same procedure...",
      "citations": [
        { "source": "CMS Medicare Claims Processing Manual",
          "section": "Chapter 23",
          "url": "https://www.cms.gov/..." }
      ] }
  ] }
```

Response — HTTP 200 (partial, RAG timeout)

When Service 3 times out, Service 2 returns detection results with explanation set to null and rag_available set to false. The frontend should show a retry option.

```
{ "session_id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "analysed",
  "rag_available": false,
  "errors": [
    { "error_id": "err_001", "module": "duplicate_charge",
      "explanation": null, "citations": [] }
  ] }
```

Response — HTTP 200 (no errors found)

```
{ "session_id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "analysed",
  "total_errors": 0,
  "total_estimated_savings": 0.00,
  "all_clear": true,
  "rag_available": true,
  "errors": [] }
```

Decisions to confirm

 error_id format — simple string like err_001 or a UUID?

Recommendation: simple string for readability in the UI

 Should confidence be a string ("high", "medium", "low") or a float (0.0–1.0)? The user stories reference both — pick one and stick to it.

Recommendation: string, since the UI only needs to render a label not a precise value

 When no errors are found, does the errors array come back empty [] or does the response include an explicit all-clear field?

Recommendation: empty array plus "all_clear": true so the frontend can distinguish "no errors found" from "analysis failed"

Endpoint 4 — POST /letter

The frontend requests dispute letter generation after the user reviews the error report. Service 2 calls Service 3 to generate letter content and returns download URLs for both Word and PDF formats.

Request

```
{ "session_id": "550e8400-e29b-41d4-a716-446655440000" }
```

Response — HTTP 200 (success)

```
{ "session_id": "550e8400-e29b-41d4-a716-446655440000",
```

```
  "status": "letter_generated",
  "downloads": {
    "docx": "https://medicheckapi.onrender.com/download/550e8400/letter.docx",
    "pdf":  "https://medicheckapi.onrender.com/download/550e8400/letter.pdf"
  } }
```

Response — HTTP 404 (no analysis results)

```
{ "error_code": "NO_ANALYSIS_RESULTS",
  "message": "No analysis results found for this session.",
  "session_id": "550e8400-e29b-41d4-a716-446655440000" }
```

Decisions to confirm

- **URL format:** Download URLs should be full absolute URLs so the frontend can call them directly without constructing the base URL.
- **File serving:** Files are served as direct download endpoints on Service 2 at MVP — no signed temporary URLs needed.

Decisions Summary

The following decisions must be agreed before coding begins. Update the Status column once each item is confirmed.

Decision	Recommendation	Status
CPT descriptions — empty or omitted?	Empty string — never populated	Agree with colleague
Confidence threshold for yellow highlight	0.80 (80%)	Agree with colleague
Bill and EOB fields — merged or separate?	Separate with source field on each line item	Agree with colleague
Confidence scores in confirmed fields?	Strip them before sending	Agree with colleague
confidence field type — string or float?	String: "high", "medium", "low"	Agree with colleague
all_clear field when no errors found?	Yes — explicit "all_clear": true	Agree with colleague
Download URLs — relative or absolute?	Full absolute URLs	Agree with colleague
error_id format	Simple string: err_001, err_002	Agree with colleague

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Error Code Reference

All error responses across all endpoints follow this consistent structure:

```
{ "error_code": "STRING",
  "message": "Human-readable description.",
  "session_id": "UUID or null" }
```

Error Code	HTTP Status	When returned
INVALID_FILE_TYPE	400	Upload is not a PDF
FILE_TOO_LARGE	400	Upload exceeds 10 MB
PAGE_LIMIT_EXCEEDED	400	Upload exceeds 20 pages
SESSION_NOT_FOUND	404	session_id does not exist
NOT_CONFIRMED	400	Analysis requested before field confirmation
NO_ANALYSIS_RESULTS	404	Letter requested before analysis
RAG_UNAVAILABLE	200	Service 3 timeout — partial response returned

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This document should be reviewed and agreed by both team members before Sprint 2 coding begins. Once agreed, commit api-contract.md to the /docs/ directory of the repository as the authoritative inter-service schema reference.